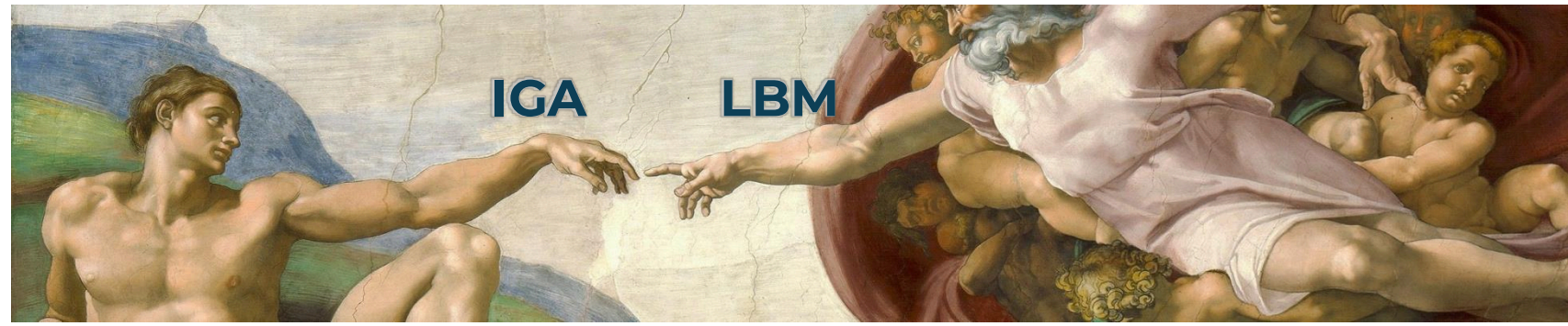




An isogeometric formulation of the lattice Boltzmann method on CAD-exact geometries

WCCM-ECCOMAS 2026 • Munich, Germany • 19–24 July 2026

MS091A: The Lattice Boltzmann method as a general-purpose tool in Computational Mechanics I

WCCM MUNICH
ECCOMAS 2026

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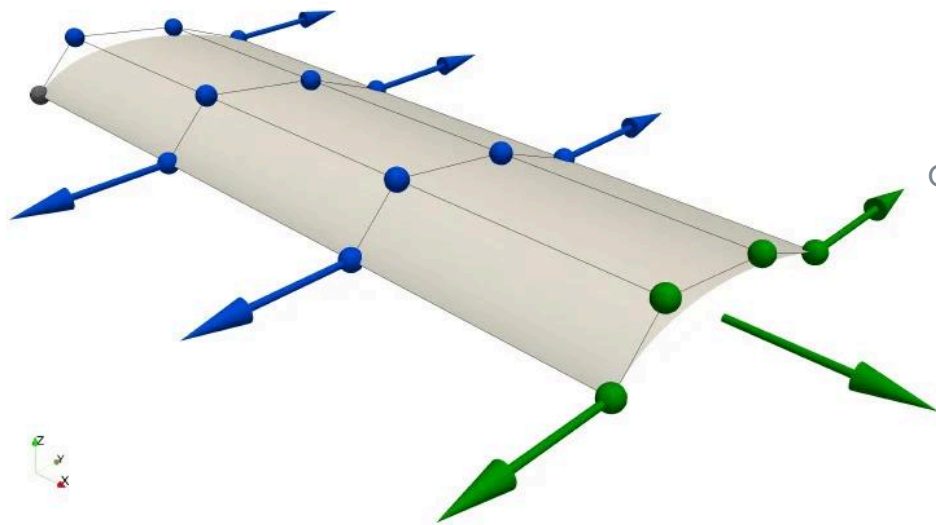


 Outline

1. Background and motivation

- 2. Isogeometric lattice Boltzmann method
- 3. Numerical examples and applications to FSI
- 4. Conclusions and outlook

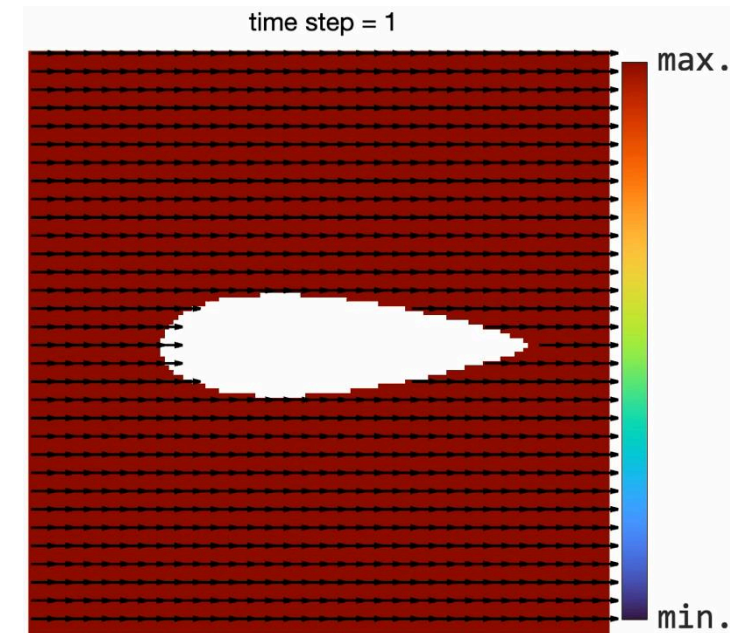
- LBM is **fast**, **scalable**, and **general-purpose**;
- But its **Cartesian grid can't represent curved geometry exactly**.



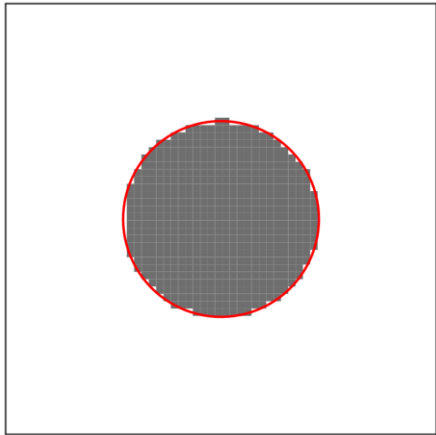
CAD-exact geometry

M. Ghommam et al. (2012). DOI: 10.4203/ccp.100.133

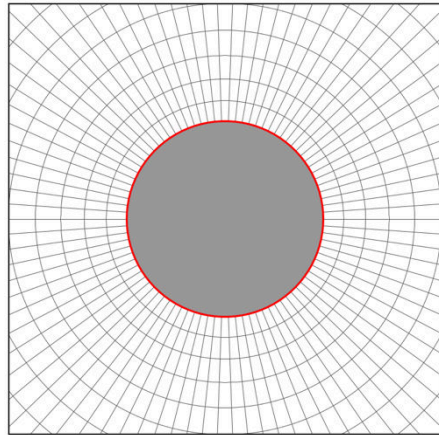
discretize with LBM



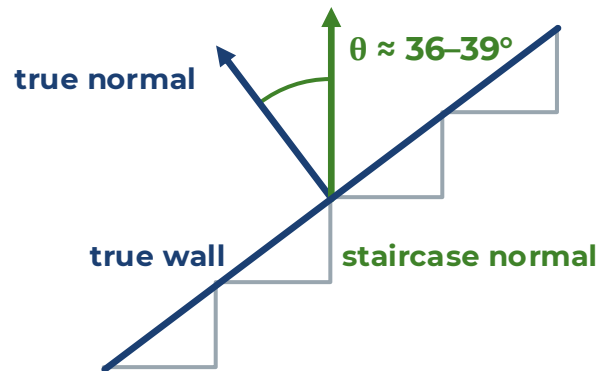
Cartesian LBM — staircased



Staircased wall

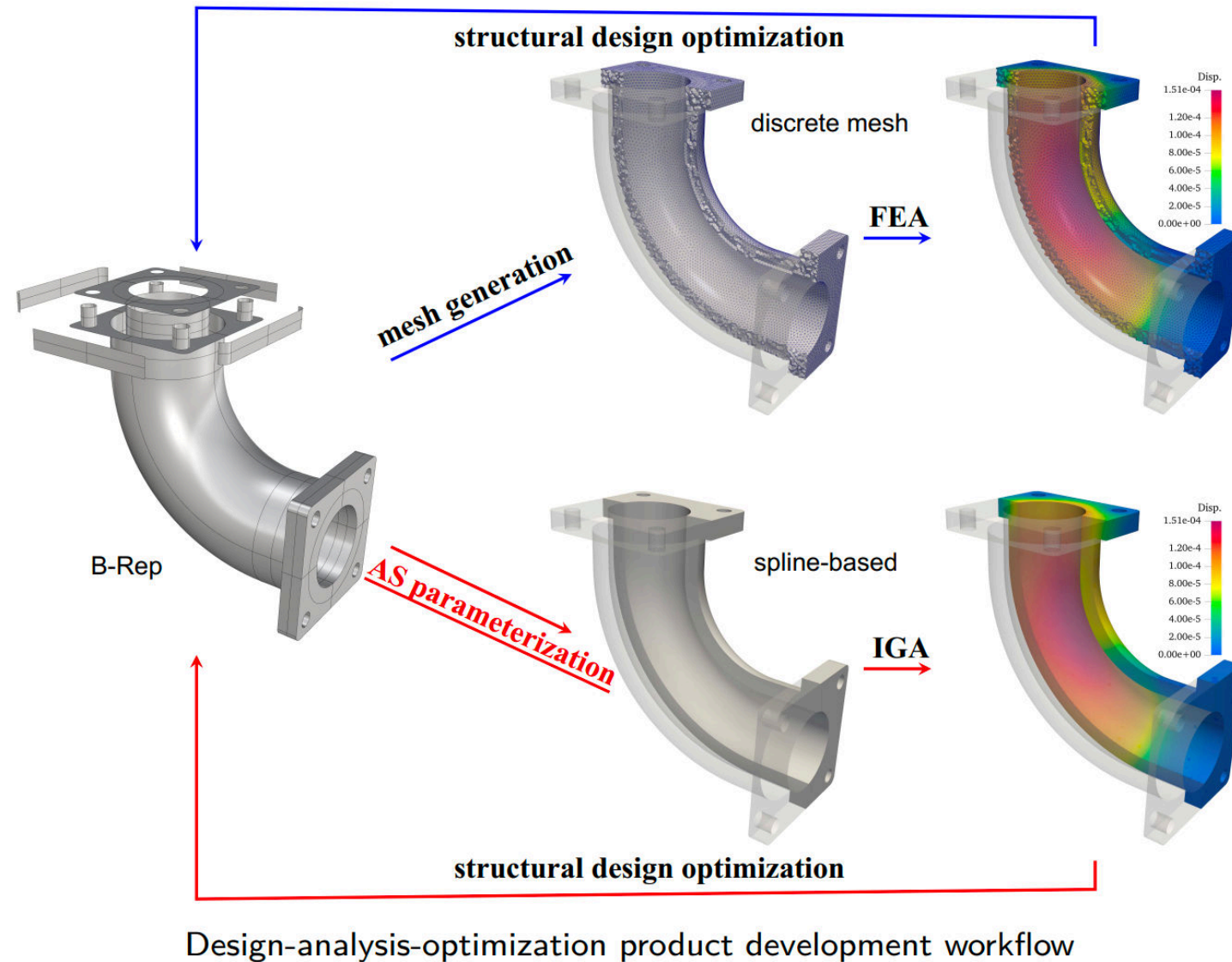


Body-fitted grid



Steps shrink with h ; facet normals stay axis-aligned

- **Exact streaming $\Rightarrow$ uniform Cartesian lattice**
 - a curved wall reduces to a staircase
- **Position error $O(h)$ — converges under refinement**
- **Normal error $36^\circ-39^\circ$ — does not converge**
 - facets remain axis-aligned at every resolution
 - spurious wall-normal velocity $\approx 0.6 U$ — no-penetration violated by $\approx 60\%$
- **Consequence: inaccurate wall** shear, pressure, drag/lift and heat transfer; spurious mass flux through the wall.



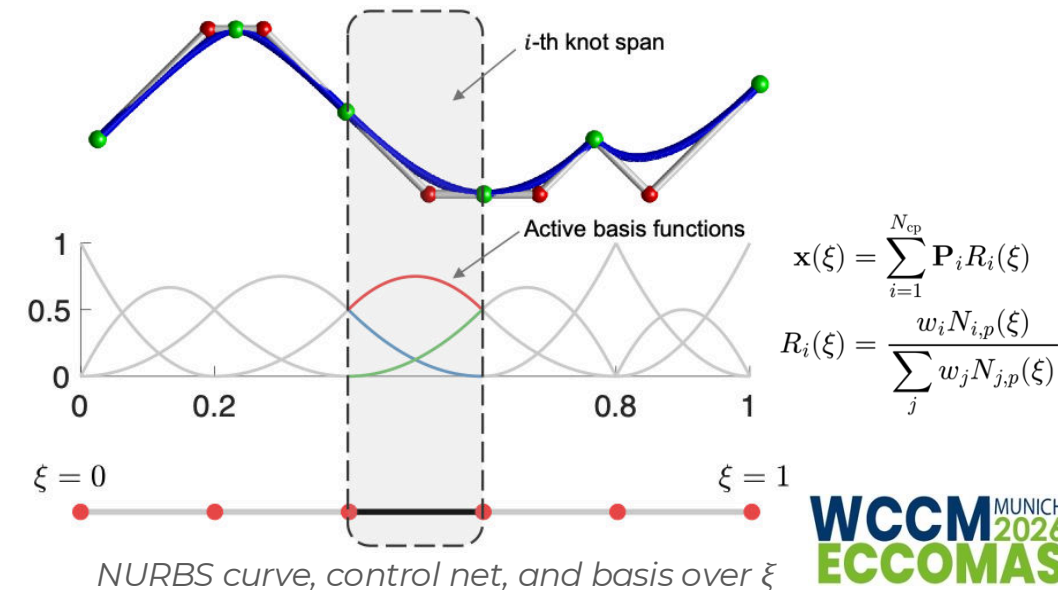
➤ **Proposed by Hughes et al., 2005.**

➤ **From CAD to analysis**

- Conventional: CAD → mesh generation → FEA
- IGA: CAD → analysis; geometry stays exact

➤ **Why it helps**

- no data conversion, exact geometry;
- design and analysis on one model;
- uniform h-, p-, k-refinement;
- high-order continuity;
- **higher accuracy per degree of freedom;**



Navier–Stokes + FVM | FEA | IGA

- + **Exact geometry representation**
 - High order / continuity approximations
 - Higher accuracy per degree of freedom
- **CAD integration → design optimisation**
- - Complex algorithm / implementation
 - High(er) assembly and solver costs
 - Difficult to achieve parallel scalability

Boltzmann + LBM

- + **Extremely simple algorithm**
- **Excellent parallel scalability**
 - Easy handling of complex geometries
 - Easy extension to multiphysics problems
- - Cartesian grids → staircase approximation
 - Poor QoI accuracy at curved boundaries
 - Stability limits at high Re
 - Weak compressibility assumption

GOAL: Keep LBM's fast, explicit solver, and replace its staircased boundaries with IGA's exact geometry.

→ IGA-LBM: the isogeometric lattice Boltzmann method.



Outline

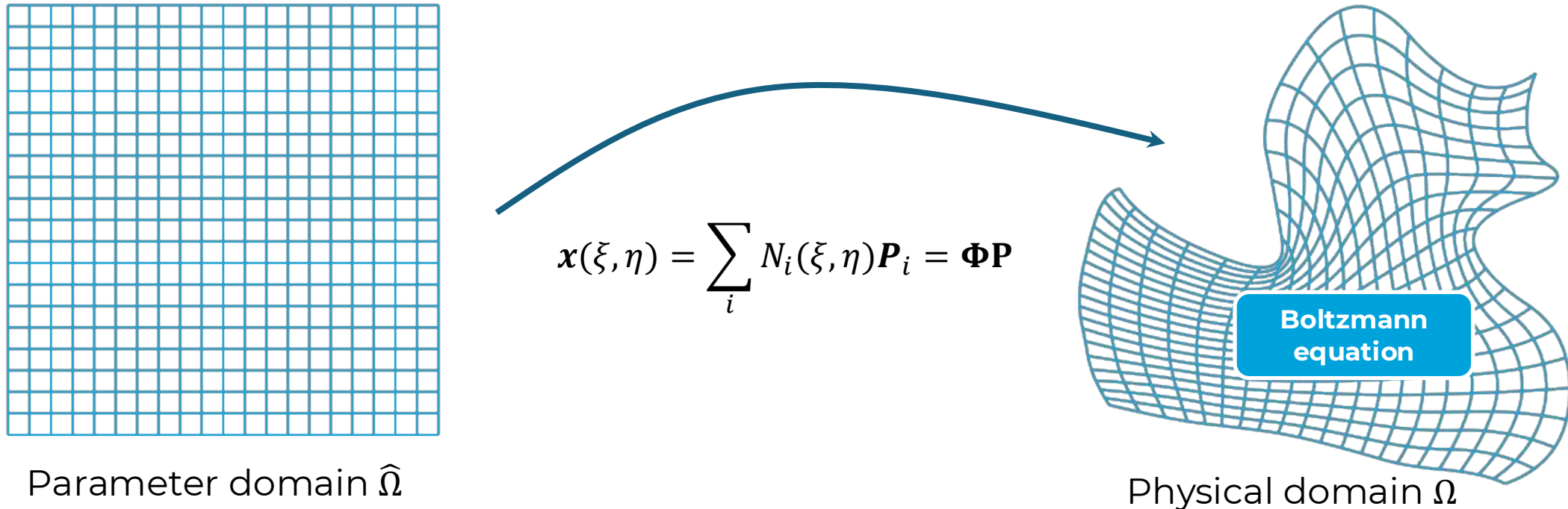
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- The physical domain Ω is parameterized by a NURBS representation $\mathbf{x}$;
- The lattice BGK equation is transformed into the parameter domain.



The relationship between the physical and computational domains satisfies

$$\begin{bmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{bmatrix} = \frac{1}{|J|} \begin{bmatrix} y_\eta & -x_\eta \\ -y_\xi & x_\xi \end{bmatrix}$$

with J being the Jacobian matrix

- Using the chain rule, the **convection term** in the **Boltzmann equation** can be rewritten as

$$\begin{aligned}
 \mathbf{e}_\alpha \cdot \nabla f_\alpha &= e_{\alpha x} \frac{\partial f_\alpha}{\partial x} + e_{\alpha y} \frac{\partial f_\alpha}{\partial y} \\
 &= e_{\alpha x} \left(\frac{\partial f_\alpha}{\partial \xi} \xi_x + \frac{\partial f_\alpha}{\partial \eta} \eta_x \right) + e_{\alpha y} \left(\frac{\partial f_\alpha}{\partial \xi} \xi_y + \frac{\partial f_\alpha}{\partial \eta} \eta_y \right) \\
 &\quad \text{can be precomputed} \rightarrow \\
 &= (e_{\alpha x} \xi_x + e_{\alpha y} \xi_y) \frac{\partial f_\alpha}{\partial \xi} + (e_{\alpha x} \eta_x + e_{\alpha y} \eta_y) \frac{\partial f_\alpha}{\partial \eta} = \tilde{e}_{\alpha \xi} \frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha \eta} \frac{\partial f_\alpha}{\partial \eta}
 \end{aligned}$$

- Represent the **particle distribution function** using NURBS representation

$$f_\alpha(\xi, \eta) = \sum_i N_i(\xi, \eta) f_{i,\alpha} = \mathbf{\Phi} \mathbf{F}_\alpha \quad \Rightarrow \quad \frac{\partial f_\alpha}{\partial \xi} = \mathbf{\Phi}_\xi \mathbf{F}_\alpha, \quad \frac{\partial f_\alpha}{\partial \eta} = \mathbf{\Phi}_\eta \mathbf{F}_\alpha$$

- Unlike IGA, LBM requires **pointwise updates** at discrete velocities. **Collocation!**

- Evaluate differential operators at the **Greville abscissae** ($\rightarrow$ Schonberg-Whiteney condition)

$$\frac{\partial f_\alpha}{\partial \xi}(\hat{\xi}, \hat{\eta}) = \hat{\Phi}_\xi \hat{\Phi}^{-1} \hat{\mathbf{F}}_\alpha, \quad \frac{\partial f_\alpha}{\partial \eta}(\hat{\xi}, \hat{\eta}) = \hat{\Phi}_\eta \hat{\Phi}^{-1} \hat{\mathbf{F}}_\alpha$$

where $\hat{\Phi}$, $\hat{\Phi}_\xi$, and $\hat{\Phi}_\eta$ are the collocation matrices and $\hat{\mathbf{F}}_\alpha$ are the collection of the particle

- ⚠ Naively computing $\hat{\Phi}_* \hat{\Phi}^{-1}$ at each time step is expensive. Precompute!

- Define $\mathbf{X}_* \hat{\Phi} = \hat{\Phi}_*$, store $\mathbf{X}_*$, and reuse in each time step

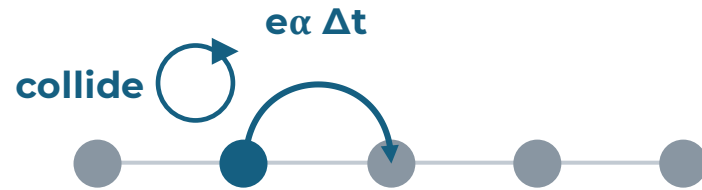
$$\hat{\Phi}^T \mathbf{X}_*^T = \hat{\Phi}_*^T \Rightarrow \mathbf{X}_*^T = (\hat{\Phi}^T)^{-1} \hat{\Phi}_*^T \Rightarrow \mathbf{X}_* = \hat{\Phi}_* \hat{\Phi}^{-1}$$

- ⚠ $\mathbf{X}_*$ does not rely on the parameterization, only on the basis functions and collocation points

Semi-discrete discrete-velocity Boltzmann equation (DVBE)

$$\frac{\partial f_\alpha}{\partial t} = S_\alpha(f) + C_\alpha(f), \quad S_\alpha(f) = -(\tilde{e}_{\alpha\xi} \mathbf{X}_\xi \hat{\mathbf{F}}_\alpha + \tilde{e}_{\alpha\eta} \mathbf{X}_\eta \hat{\mathbf{F}}_\alpha), \quad C_\alpha(f) = -\frac{1}{\tau} (f_\alpha - f_\alpha^{eq})$$

Classical LBM: stream–collide



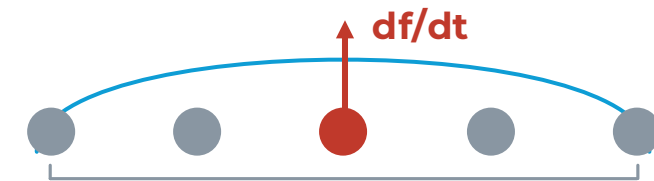
collide (relax), then stream

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \Delta t, t + \Delta t) = f_\alpha(\mathbf{x}, t) - \frac{\Delta t}{\tau} (f_\alpha - f_\alpha^{\text{eq}})$$

fused collide + stream in one lattice step

- packets hop along fixed links
- space–time locked $\Rightarrow$ uniform lattice only
- $\mathbf{v} = \mathbf{c}_s^2(\tau - \frac{1}{2}\Delta t)$ — time-step artefact
- curved walls $\Rightarrow$ staircased bounce-back

This work: collocation method of lines

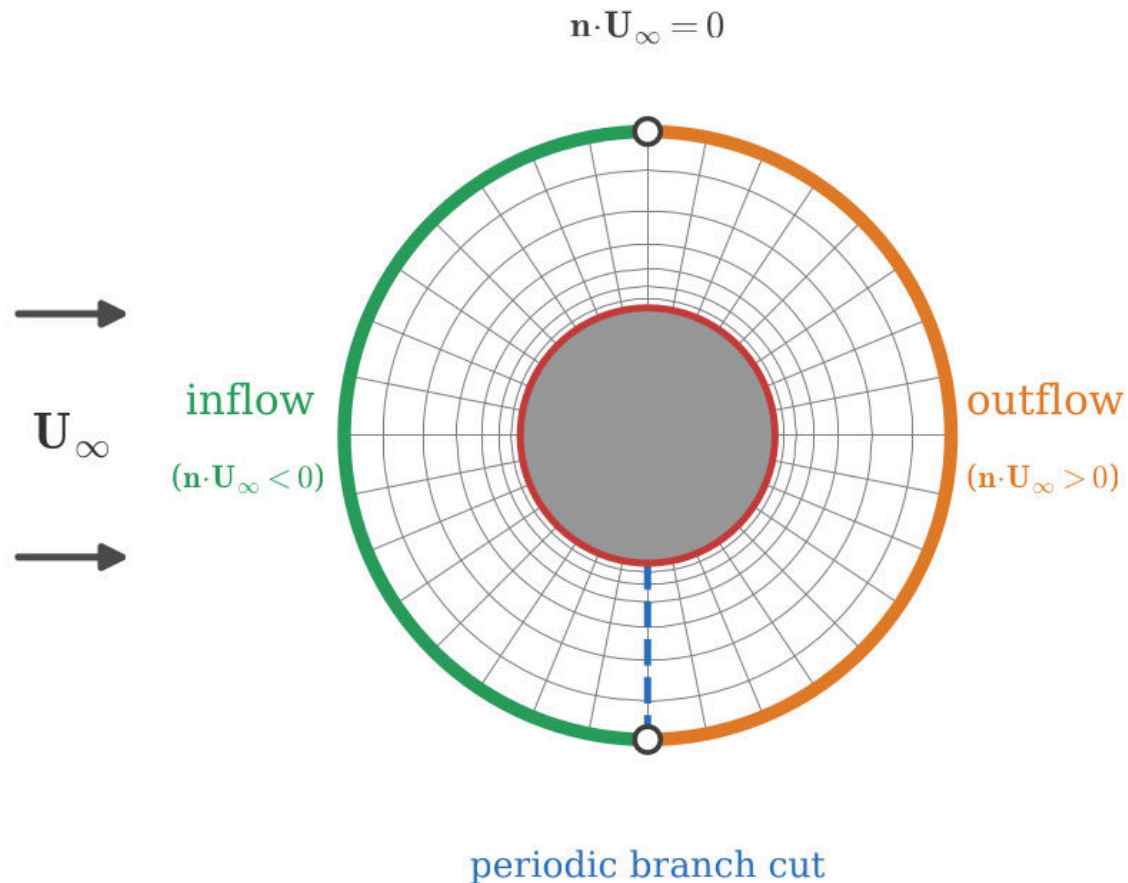


collocation derivative $D\xi$ (banded 5-point)

$$\frac{\partial f_\alpha}{\partial t} + \underbrace{\tilde{e}_{\alpha\xi} \frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha\eta} \frac{\partial f_\alpha}{\partial \eta}}_{\text{transformed advection}} = -\frac{1}{\tau} (f_\alpha - f_\alpha^{\text{eq}})$$

$$\tilde{e}_\alpha = J^{-1} \mathbf{e}_\alpha \text{ (contravariant, precomputed once)}$$

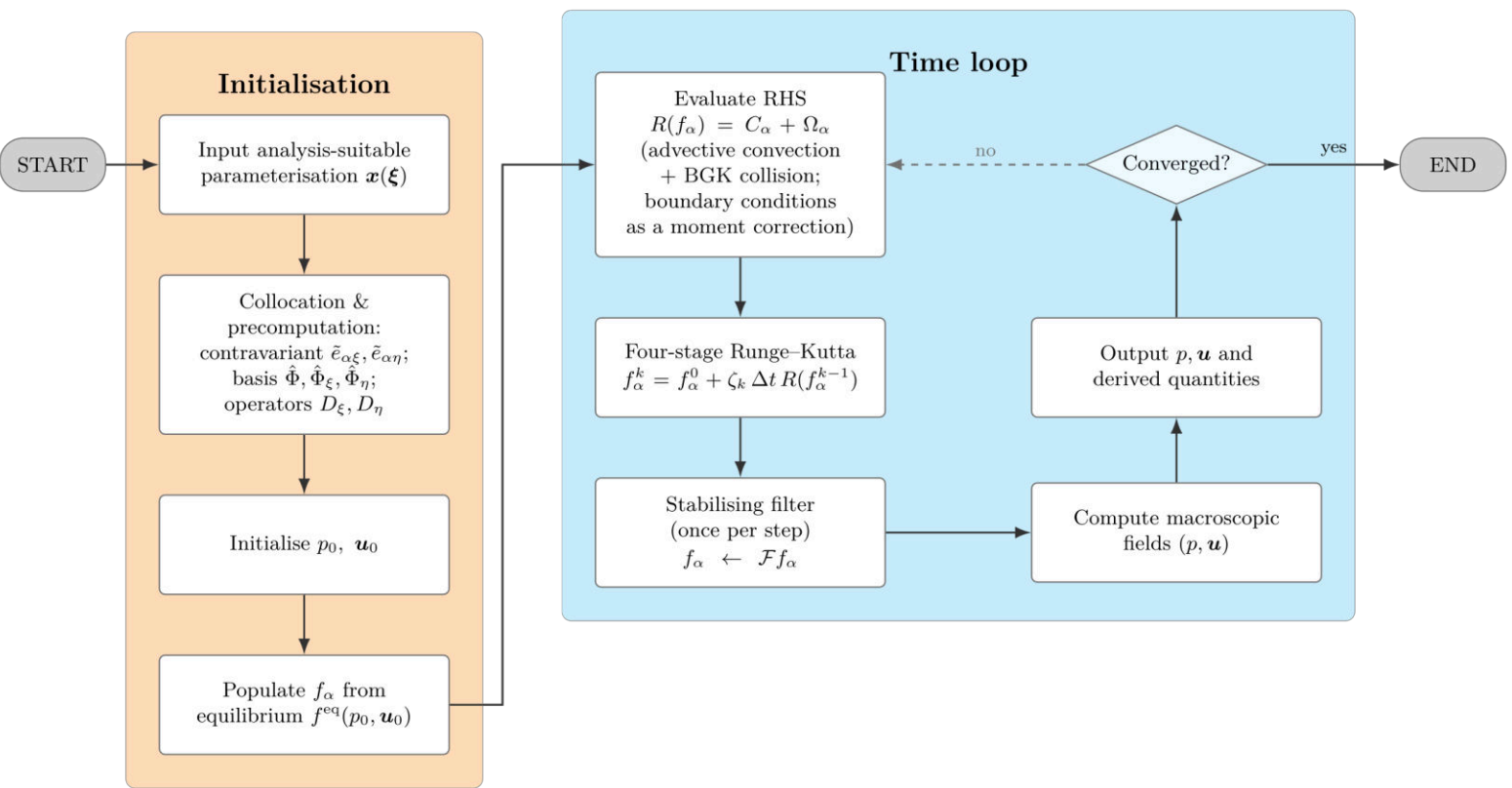
- nodal spline values evolve in place
- no Lagrangian streaming
- continuous-time BGK $\rightarrow$ RK4 + low-pass filter
- $\mathbf{v} = \mathbf{c}_s^2 \tau$ — no time-step artefact



➤ Impose at the moment level: prescribe $u, p \rightarrow$ equilibrium target (f_α never overwritten).

- **Wall (no-slip):** $u = u_{\text{wall}}, \partial p / \partial n = 0$
- **Inflow (Dirichlet):** $u = U_\infty, p = p_\infty$
- **Outflow (Neumann):** $\partial u / \partial n = 0$
- **Branch cut (periodic):** $f(\theta = 0) = f(\theta = 2\pi)$

$$u(x_b) = u_{\text{wall}}, \quad \left. \frac{\partial p}{\partial n} \right|_{x_b} = 0, \quad f_\alpha^{\text{eq}}(x_b) = f_\alpha^{\text{eq}}(p_b, u_{\text{wall}})$$



Per step

- 4 RK stages + 1 filter
- BCs enter the RHS at every stage

Cost

- local B-spline support $\Rightarrow$ banded operators $\Rightarrow O(N)$ per step
- **0.36–0.38 μs / DOF / step**
- $\approx$ a few classic LBM updates

Time step

- **Δt free, unlike classic LBM**
- collision sets Δt , not flow speed
- use $\Delta t = \min\{2\tau, CFL/s_{\max}\}$, $CFL \approx 0.1\text{--}0.3$

$$\frac{df_\alpha}{dt} = R_\alpha(f) := \underbrace{C_\alpha}_{\text{advection}} + \underbrace{\Omega_\alpha}_{\text{BGK}}$$

Semi-discrete transport

$$f_\alpha^{(s)} = f_\alpha^n + \zeta_s \Delta t R_\alpha(f^{(s-1)}), \quad \zeta_s = \left(\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1\right)$$

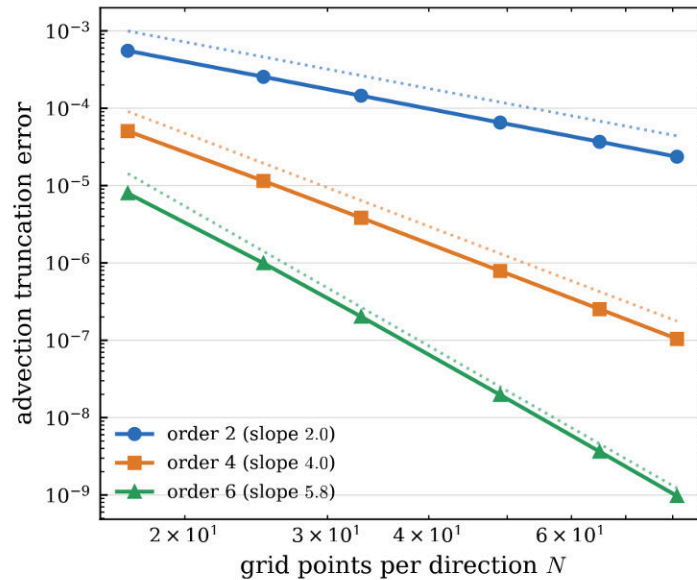
Low-storage RK4 · 4 stages

$$\Delta t \leq \min\left\{2.785 \tau, CFL/s_{\max}\right\}, \quad \tau = \mathcal{O}(\text{Re}^{-1})$$

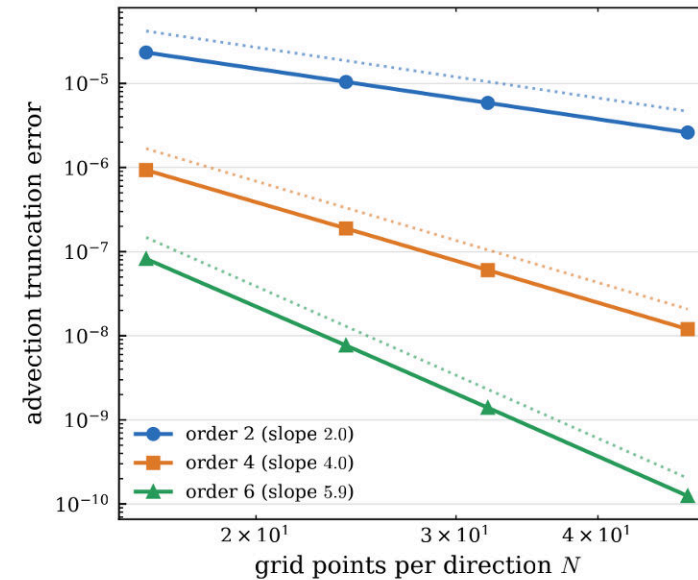
Explicit stability bound

 Outline

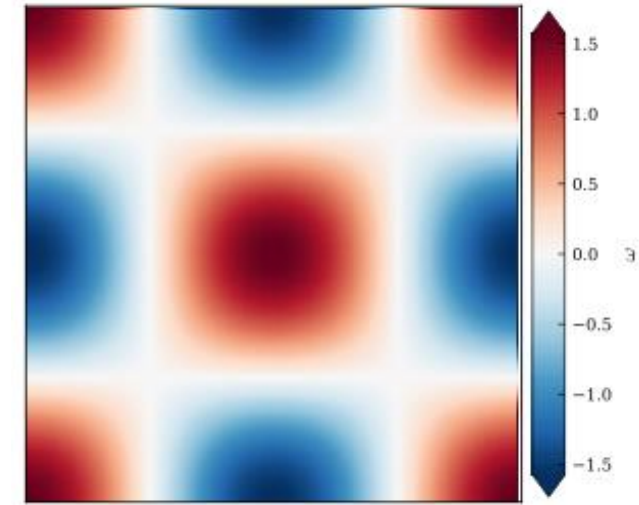
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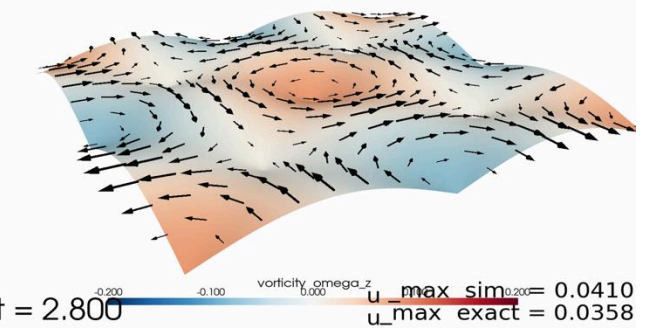
2D Taylor–Green vortex



3D ABC / Beltrami flow (D3Q19)

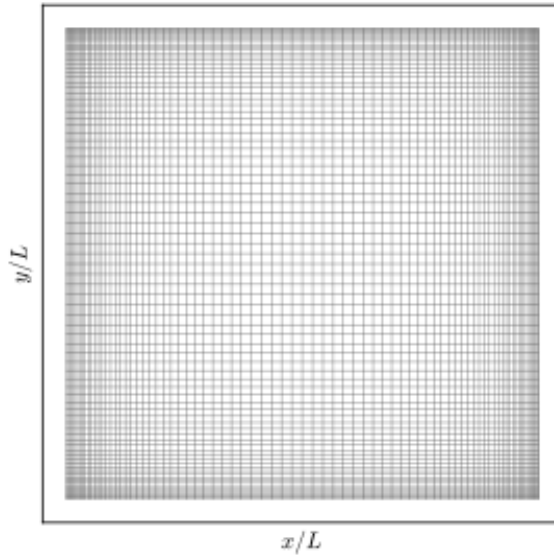


IGA-LBM 2D Taylor–Green Vortex
 $N_x = 64, N_y = 64$ $\tau = 0.550$ $\nu = 0.1833$

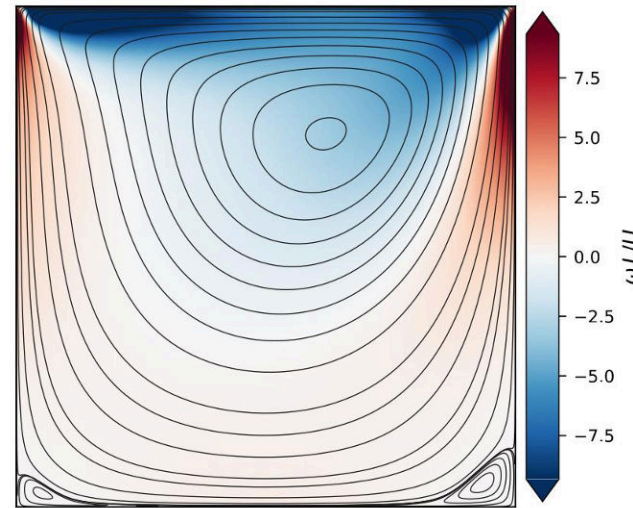


2D Taylor–Green vortex

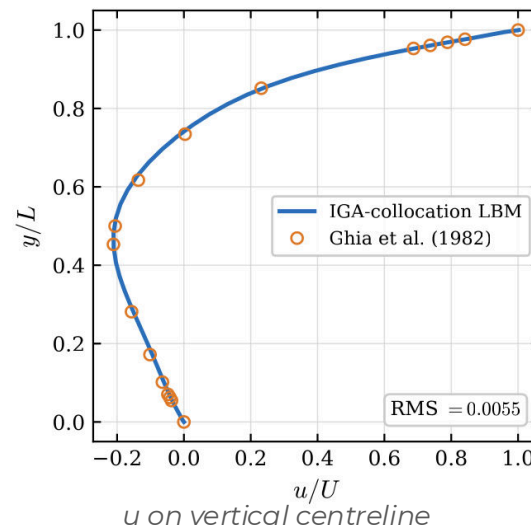
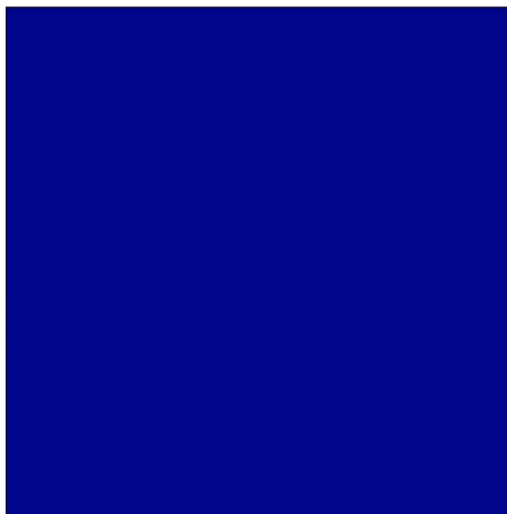
- **Exact-solution benchmarks:** Taylor–Green vortex and 3D Arnold–Beltrami–Childress flow flow: error measured against truth.
- **Design orders recovered:** measured slopes hit 2 / 4 / 6 in every case.
- **No loss at the wall.**



Clustered grid



vorticity + streamlines



Setup: 48×48 stretched grid $\cdot Re = 100$

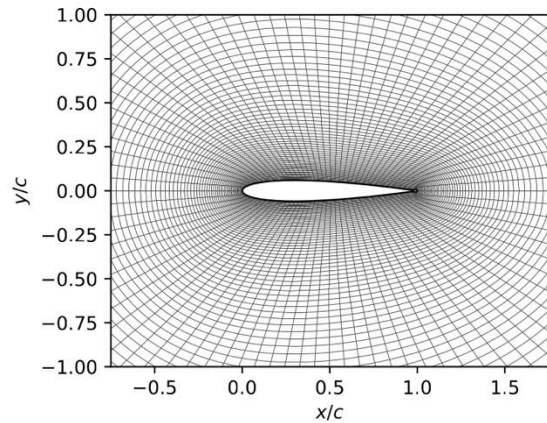
- Square domain

Benchmark: Ghia et al. (1982)

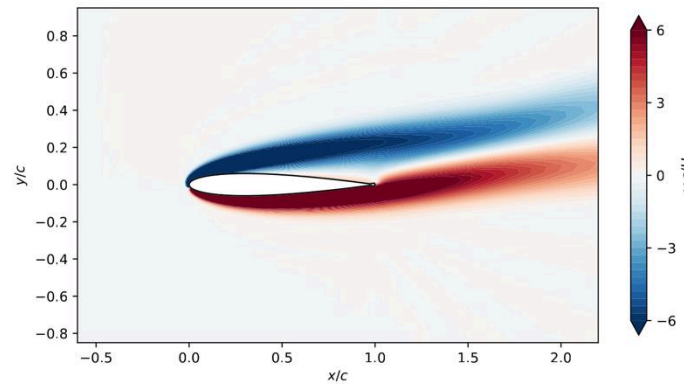
- Centreline velocity matches: $RMS\ 5.5 \times 10^{-3}$ (u), 4.3×10^{-3} (v)

Grid K×K	24	32	48	64
RMS vs Ghia	1.92×10^{-2}	1.11×10^{-2}	5.50×10^{-3}	5.50×10^{-3}

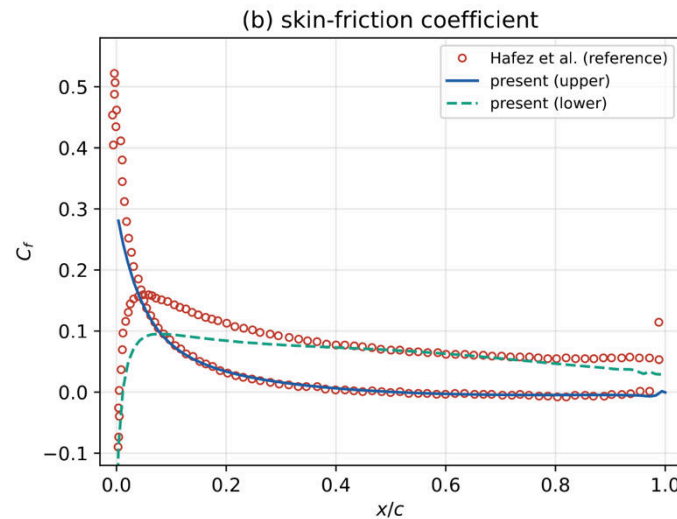
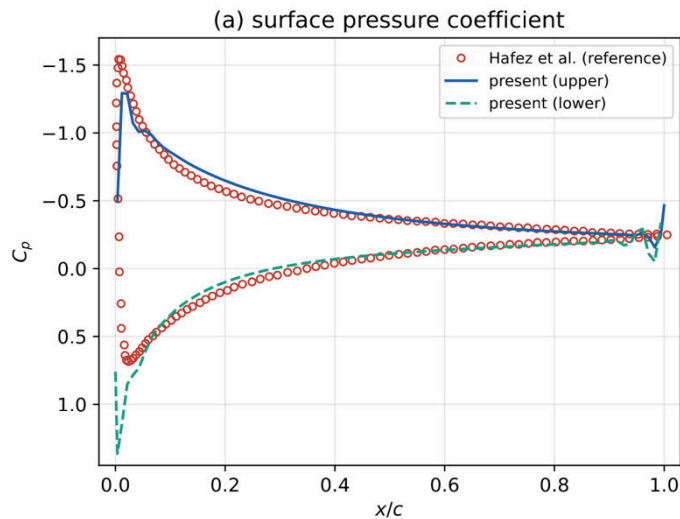
- Refine the mesh $\rightarrow$ error falls, then flattens by 48^2
- That plateau is the $O(Ma^2)$ compressibility floor: the athermal-LBM model limit, not the mesh
- So high order reaches the floor on a very coarse grid



elliptic body-fitted O-grid (180x110)



vorticity: LE suction, shear layers, thin wake



surface pressure and skin friction vs Hafez et al. (2006)

➤ Setup: NACA0012 • $Re = 500$ • $\alpha = 10^\circ$

- Body-fitted O-grid 180x110

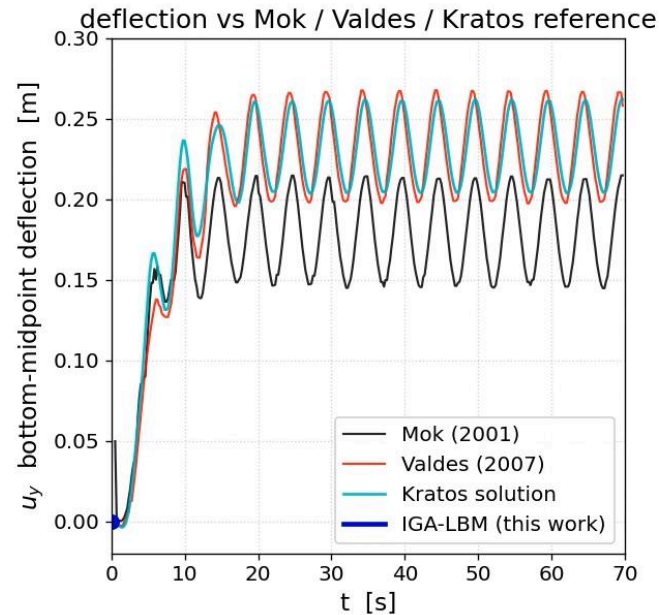
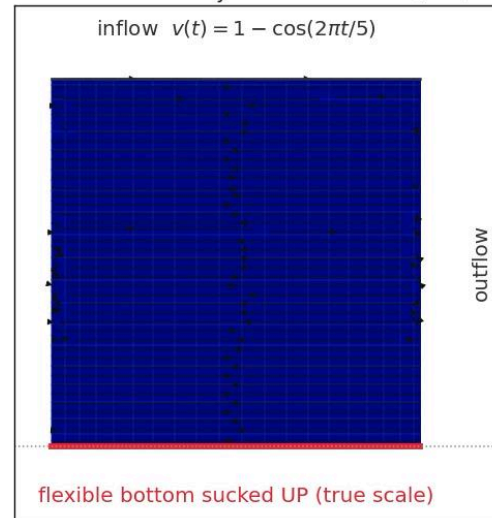
➤ Real CAD grid

- Grid generated by an external mesher, only sampled coordinates (no analytic map)

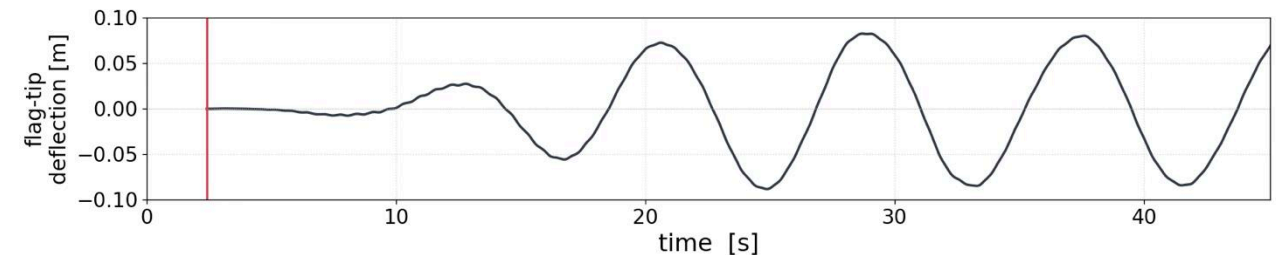
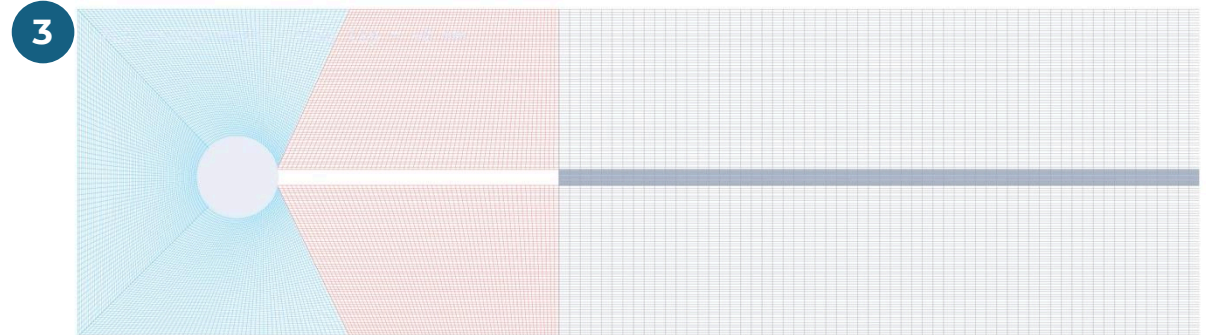
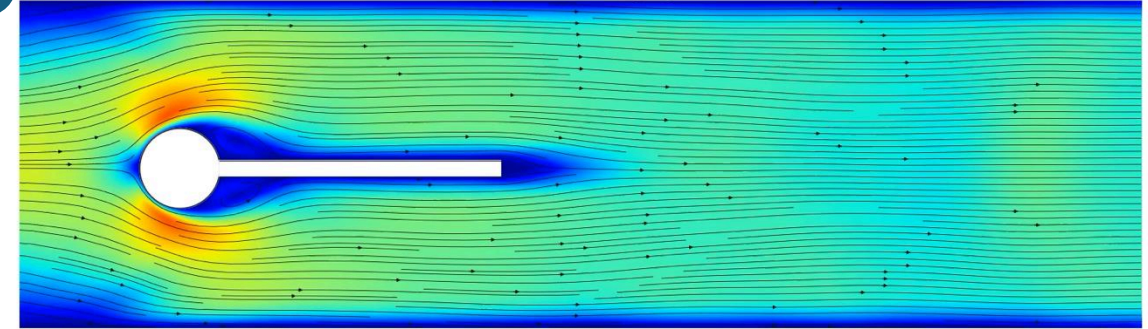
➤ It works

- **Uniform flow exact to 7×10^{-15}**
- Forces match references: $C_l = 0.295$, $C_d = 0.134$
- Ref. Hafez et al. (2006)

1 Mok-Wall FSI lid-driven cavity $t = 0.0$ s $u_y = +0.000$ m (UP)



2 Turek FSI2 $t = 2.4$ flag tip = +0 mm



Moving boundaries on a body-fitted grid — same solver.

- 1 Mok flexible wall** — panel deflecting in channel flow; FSI verification case.
- 2 Turek-Hron FSI2** — elastic flag oscillating behind a fixed cylinder.
- 3 Body-fitted mesh motion** — grid deforms with the structure; no remeshing.



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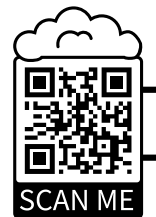
Conclusions

- **Body-fitted isogeometric LBM:** strong-form BGK on NURBS, RK4, high-order filter.
- **Geometry exactness:** high order accuracy holds right up to the wall.
- **Validated & general:** TGV, lid-driven cavity, NACA0012; also 3D and FSI.

Outlook

- **Complex multipatch CAD:** extend and apply to trimmed, multipatch geometries.
- **Stiff-accurate IMEX:** lift the explicit $\Delta t = O(\text{Re}^{-1})$ limit.
- **GPU / matrix-free:** body-fitted 3D and moving-boundary FSI.

Preprint: <https://arxiv.org/pdf/2509.11427v2>

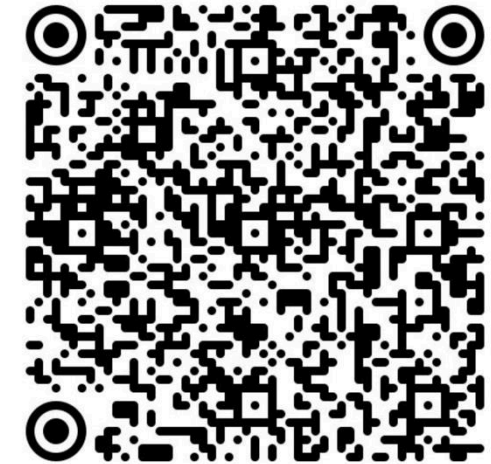


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Many thanks for your kind attention!

Q&A

If you are interested in my research, please feel free to contact me! ;-)

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